



Contact

Phone

+40733646257

Email

catalindogaru22@gmail.com

Address

10 Corneliu Coposu Blvd., Timișoara

SKILLS

- Python, Java, C/C++, SQL
- Machine Learning, Deep Learning (LSTM), Computer Vision, OpenCV, scikit-learn, MediaPipe
- PostgreSQL, Flyway, REST APIs
- Git, GitHub, Jira, Agile/Scrum

LANGUAGES

- Romanian: Native
- English: Advanced
- Russian: Fluent

Dogaru Cătălin

COMPUTER SCIENCE & INFORMATION
TECHNOLOGY STUDENT

EXPERIENCE

04/2026 - PRESENT

MIND AUTOMOBILE PARTS ROMANIA | IT Intern

Timișoara, România

- **Device Configuration and Technical Support:** Configuring, and troubleshooting hardware and IT infrastructure across the company, providing rapid technical support to ensure business continuity.
- **Proactive Maintenance and System Administration:** Ensuring the availability and security of internal platforms through regular maintenance
- **Process Automation:** Collaborating with various departments to identify manual workflows and participating in the development of targeted automation solutions to improve efficiency.

05/2026 - PRESENT

AC LABS (ACCESS) | AI Trainee (Program: Powered by AI, Driven by You)

Timișoara, România

- **Development of an Intelligent Automation Platform:** Contributing to the construction of QuizMaster, an end-to-end web application for automated content management, using React and .NET 10 Web API.
- **Process Optimization through AI:** Implementing LLM models (Azure OpenAI) for automated content generation and data summarization, significantly reducing manual operational effort.
- **Implementation of Semantic Search Technologies:** Developing modules for data vectorization and integrating Vector Databases, optimizing search precision and the analysis of complex information.
- **Software Architecture and Efficiency:** Applying SOLID principles, Clean Architecture, and Design Patterns to ensure a scalable, high-performance, and easily automatable platform.
- **Utilization of AI Programming Tools:** Increasing development productivity by leveraging AI assistants such as Claude Code and GitHub Copilot, maintaining professional standards for code review and collaborative teamwork.

RELEVANT PROJECTS

11/2025 - PRESENT

AI Personal Trainer & Exercise Recognition System

Personal Project

- **Project Description:**
End-to-end application (Computer Vision & ML) acting as a real-time fitness coach.
- **Key Technical Achievements:**
 - **AI Recognition:** Trained a Deep Learning (LSTM) model to dynamically recognize exercise types and automatically switch training modes.
 - **Biomechanical Calculation:** Created functions based on 3D coordinates to measure joint angles, validating only repetitions with a full range of motion.
 - **Real-Time Feedback:** Implemented a system that compares execution against a reference dataset, providing live visual alerts via OpenCV and real-time auditory coaching to cancel incorrect reps and count successful ones.
 - **Spatial Optimization:** Normalized coordinates relative to the hip axis, ensuring AI accuracy regardless of the user's distance from the camera (1-3 meters).

- Technologies:
Python, PyTorch, MediaPipe, OpenCV, Deep Learning (LSTM), Computer Vision, Biomechanical Modeling

05/2025 - PRESENT

ML Platform for Cryptocurrency Analysis

Team Project

- **Project Description:**
Scalable Python backend for cryptocurrency price analysis and forecasting using Machine Learning, developed in an Agile environment.
- **Key Technical Achievements:**
 - Development & Data: Developed modular architecture to ingest and normalize live data from the CoinGecko API, implementing advanced feature engineering to optimize model training.
 - Machine Learning & Automation: Collaborated on training predictive models and developed an automated system for sending daily price predictions via email.
 - DevOps & Collaboration: Managed workflow using Jira (Sprints) and GitHub, ensuring code quality through Unit & Integration testing.
- **Technologies:**
Python, scikit-learn, pandas, PostgreSQL, Flyway, REST API Integration, Git/GitHub, Jira, Agile/Scrum

EDUCATION

2023 - Present

Politehnica University of Timișoara

Bachelor's Degree in Computer Science and Information Technology

2020 - 2023

"Ion Creangă" Theoretical High School, Chișinău, Moldova

Baccalaureate Diploma